When does multivariate forecast reconciliation help?

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Abstract

When several variables are observed over the same hierarchy, it seems natural to reconcile their forecasts jointly so that cross-variable dependence can inform the adjustment. We ask when joint reconciliation improves on reconciling each variable separately. Writing minimum trace reconciliation as a regression of the base forecast errors on the observed incoherences, we show that joint and separate reconciliation coincide if and only if each variable's separately reconciled errors are uncorrelated with the other variables' incoherences. This holds, for example, when the error covariance is separable or when the cross-variable dependence is confined to the coherent part of the errors. We propose two measures and a likelihood ratio test of this condition. In simulations, joint reconciliation reduces the mean squared error by at most 15.7% even when the true error covariance is known, and with an estimated covariance it is less accurate than separate reconciliation in small samples. We compare joint and separate reconciliation to monthly Brazilian employment admissions and dismissals over 2004–2019. With ARIMA base forecasts, joint reconciliation improves the forecasts of dismissals but not of admissions; with VAR base forecasts, it is worse than separate reconciliation for both. For net employment change, the difference between admissions and dismissals, joint reconciliation of the ARIMA forecasts reduces the national mean squared error by 17%. For probabilistic forecasts of net change, simulating the base forecasts jointly reduces the continuous ranked probability score (CRPS) by 3%, and reconciling them jointly as well raises the reduction to 12%. Separate reconciliation is often sufficient for forecasting each variable, but the dependence between variables should be modelled when the quantity of interest combines them.

Keywords: hierarchical time series, coherent forecasts, minimum trace, Kronecker product, probabilistic forecasting, labour market

1. Introduction

Collections of time series can often be aggregated at several levels by geography, product, or other attributes. Forecasts are usually needed for every series at every level, and they should be coherent: each aggregate forecast should equal the sum of the forecasts below it. Forecast reconciliation achieves this by forecasting every series independently and then adjusting the forecasts to satisfy the aggregation constraints (Hyndman et al., 2011; Athanasopoulos et al., 2024). The minimum trace (MinT) method of Wickramasuriya et al. (2019) chooses the adjustment using the covariance matrix of the base forecast errors, and it is now the standard approach.

Reconciliation is almost always applied to one variable at a time, yet many hierarchies carry several variables. A retailer, for example, may need forecasts of both unit sales and revenue for every store, with each variable summing across stores to regional and national totals. The two variables are

measured in different units, but both respond to the same promotions, holidays and local demand, so their forecast errors are correlated. Because MinT works by exploiting the covariance of the base forecast errors, it seems natural to reconcile both variables together, using a covariance matrix that spans them, and to expect this to beat reconciling each separately. We ask when joint reconciliation is better than reconciling each variable separately.

Joint reconciliation of several variables is not the same problem as reconciling a grouped hierarchy. In a grouped hierarchy, such as sales disaggregated by both region and product type, the categories of each grouping variable are linked by aggregation constraints: total sales is the sum over products or over regions, and that total is itself forecast. In the retail example, by contrast, unit sales are never added to revenue. The stacked summing matrix is block diagonal by variable, and the only link between the variables is the error covariance.

Some cross-variable dependence is already known to be of no use to reconciliation. MinT is unaffected by any component of the error covariance that lies in the coherent subspace (Wickramasuriya, 2021), so cross-variable dependence of that form is invisible. When the error covariance is separable (i.e., a Kronecker product of a cross-variable matrix and a cross-series matrix), joint and separate reconciliation coincide whatever the cross-variable correlation. This follows from a result of Girolimetto and Di Fonzo (2025) for cross-temporal reconciliation, and it is the reconciliation analogue of Zellner's observation (Zellner, 1962) that seemingly unrelated regressions with identical regressors reduce to equation-by-equation least squares. These conditions are sufficient, but they do not say which forms of dependence joint reconciliation can exploit, or by how much.

Our approach is to write MinT as a regression of the base forecast errors on the observed incoherences of the base forecasts. In this form, the question becomes whether one variable's incoherences carry information about another variable's errors beyond what that variable's own incoherences provide. This leads to a necessary and sufficient condition for joint and separate reconciliation to coincide, which includes the known cases above.

In practice, we need to know whether this condition holds for a given data set, and if not, how much the departure affects the forecasts. We propose two measures of the departure from this condition: κ, the weight that the joint reconciliation places on other variables' incoherences, and γ, the reduction in mean squared error that it would deliver if the estimated covariance were correct. We also propose a likelihood ratio test of the condition, which uses the regression form of MinT and needs no simulation.

We study the effect of joint reconciliation in two simulation experiments, one in which the error covariance is controlled directly and one in which base forecasting models are fitted to simulated series. We also explore joint reconciliation in an application to monthly worker admissions and dismissals in Brazil's 27 federative units, which aggregate to five regions and to the country. Besides the forecasts of each variable, we consider quantities that combine the variables. Net employment change, the difference between admissions and dismissals, is the headline figure in Brazil's official monthly employment statistics. Its forecast distribution depends on the joint distribution of the base forecasts, so for probabilistic forecasts we ask whether the dependence between the variables should be modelled when the base forecasts are simulated, when they are reconciled, or both.

Section 2 sets out notation and the reconciliation problem for several variables. Section 3 gives necessary and sufficient conditions for joint and separate reconciliation to coincide, and their consequences. Section 4 develops the measures and the test. Section 5 reports the two experiments, Section 6 the application to Brazilian employment, and Section 7 draws out the implications.

2. Reconciling several variables

2.1. Notation

We follow the notation of Athanasopoulos et al. (2024), adapted to several variables. Suppose m variables are each observed over the same hierarchy with n series, of which n_b are at the bottom level

and $n_a = n - n_b$ are aggregates. For variable j, let $\boldsymbol{y}_{j,t}$ be the n-vector of all its series at time t, and $\boldsymbol{b}_{j,t}$ the n_b-vector of its bottom-level series. Then

$$\boldsymbol{y}_{j,t} = \boldsymbol{S}\boldsymbol{b}_{j,t}, \qquad \boldsymbol{S} = \begin{bmatrix} \boldsymbol{A} \\ \boldsymbol{I}_{n_b} \end{bmatrix}, \tag{1}$$

where the $(n_a \times n_b)$ matrix $\boldsymbol{A}$ gives the aggregation constraints. Equivalently, $\boldsymbol{C}\boldsymbol{y}_{j,t} = \boldsymbol{0}$ with $\boldsymbol{C} = [\boldsymbol{I}_{n_a} \quad -\boldsymbol{A}]$, so that $\boldsymbol{C}\boldsymbol{S} = \boldsymbol{0}$. Figure 1 shows a three-level example with $n = 8$ and $n_b = 5$.

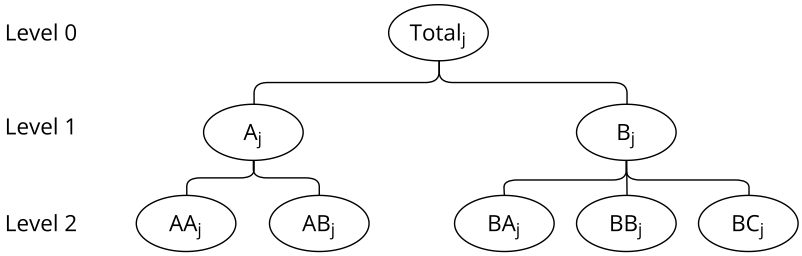


Figure 1: A hierarchy with three levels, observed for each of m variables.

Collect the variables in $\boldsymbol{Y}_t = [\boldsymbol{y}_{1,t}, \ldots, \boldsymbol{y}_{m,t}]$ and $\boldsymbol{B}_t = [\boldsymbol{b}_{1,t}, \ldots, \boldsymbol{b}_{m,t}]$. Stacking the columns gives

$$\operatorname{vec}(\boldsymbol{Y}_t) = \boldsymbol{S}_* \operatorname{vec}(\boldsymbol{B}_t), \qquad \boldsymbol{S}_* = \boldsymbol{I}_m \otimes \boldsymbol{S}, \tag{2}$$

with constraint matrix $\boldsymbol{C}_* = \boldsymbol{I}_m \otimes \boldsymbol{C}$. The stacked summing matrix is block diagonal: no constraint links one variable to another. This distinguishes the setting from a grouped hierarchy, where the categories of a grouping variable (such as travel purpose in the Australian tourism data of Wickramasuriya et al. (2019)) are linked by aggregation constraints because their total is itself observed and forecast.

2.2. *Forecast reconciliation*

Let $\widehat{\boldsymbol{y}}_h$ be the stacked base forecasts of all mn series at time $T + h$, given data up to time T and produced without regard to the constraints, and let $\boldsymbol{W}_h$ be the covariance matrix of their errors $\boldsymbol{y}_{T+h} - \widehat{\boldsymbol{y}}_h$. Linear reconciliation maps the base forecasts to coherent forecasts $\widetilde{\boldsymbol{y}}_h = \boldsymbol{S}_*\boldsymbol{G}\widehat{\boldsymbol{y}}_h$ for some matrix $\boldsymbol{G}$ with $\boldsymbol{G}\boldsymbol{S}_* = \boldsymbol{I}$. The minimum trace (MinT) choice (Wickramasuriya et al., 2019) minimises the trace of the reconciled error covariance and gives

$$\widetilde{\boldsymbol{y}}_h = \boldsymbol{S}_*(\boldsymbol{S}_*'\boldsymbol{W}_h^{-1}\boldsymbol{S}_*)^{-1}\boldsymbol{S}_*'\boldsymbol{W}_h^{-1}\widehat{\boldsymbol{y}}_h = \left[\boldsymbol{I} - \boldsymbol{W}_h\boldsymbol{C}_*'(\boldsymbol{C}_*\boldsymbol{W}_h\boldsymbol{C}_*')^{-1}\boldsymbol{C}_*\right]\widehat{\boldsymbol{y}}_h, \tag{3}$$

the second form being the zero-constrained representation of Di Fonzo and Girolimetto (2023). Applied to the stacked system this is ordinary MinT with summing matrix $\boldsymbol{S}_*$; we call it *joint* reconciliation. *Separate* reconciliation applies MinT to each variable on its own, using only the diagonal block of $\boldsymbol{W}_h$ for that variable. Joint reconciliation can use the cross-variable blocks of $\boldsymbol{W}_h$, and it is natural to expect it to be more accurate when the variables' forecast errors are correlated. Section 3 shows when that expectation holds.

In practice $\boldsymbol{W}_h$ is unknown. Following Wickramasuriya et al. (2019), we set $\widehat{\boldsymbol{W}}_h = k_h\widehat{\boldsymbol{W}}_1$, where $\widehat{\boldsymbol{W}}_1$ is the shrinkage estimator

$$\widehat{\boldsymbol{W}}_1 = \lambda\widehat{\boldsymbol{W}}_{1,D} + (1 - \lambda)\widehat{\boldsymbol{W}}_{1,S}, \tag{4}$$

in which $\widehat{\boldsymbol{W}}_{1,S}$ is the sample covariance matrix of the one-step in-sample residuals, computed about zero, $\widehat{\boldsymbol{W}}_{1,D}$ is its diagonal, and the intensity λ is estimated as in Schäfer and Strimmer (2005). The scale k_h cancels from (3). For separate reconciliation the estimator is applied to each variable's residuals on its own, so each variable has its own shrinkage intensity.

3. When does joint reconciliation matter?

3.1. Joint and separate reconciliation

Stack the m variables as in (2), with summing matrix $\boldsymbol{S}_*$ and constraint matrix $\boldsymbol{C}_*$. Let $\boldsymbol{W}$ be the $(mn \times mn)$ covariance matrix of the stacked base forecast errors $\boldsymbol{e} = (\boldsymbol{e}_1', \dots, \boldsymbol{e}_m')'$, where $\boldsymbol{e}_j$ holds the n errors of variable j, and write $\boldsymbol{W}_{jk} = \mathrm{Cov}(\boldsymbol{e}_j, \boldsymbol{e}_k)$ for its $(n \times n)$ blocks. We drop the horizon subscript h throughout this section.

Joint MinT reconciliation in zero-constrained form (Di Fonzo and Girolimetto, 2023) is $\widetilde{\boldsymbol{y}} = \boldsymbol{M}\widehat{\boldsymbol{y}}$ with

$$\boldsymbol{M} = \boldsymbol{I} - \boldsymbol{L}\boldsymbol{C}_*, \qquad \boldsymbol{L} = \boldsymbol{W}\boldsymbol{C}_*'(\boldsymbol{C}_*\boldsymbol{W}\boldsymbol{C}_*')^{-1}. \tag{5}$$

Separate reconciliation applies univariate MinT to each variable using its own error covariance $\boldsymbol{W}_{jj}$. Its map is block diagonal, $\boldsymbol{M}^{\text{sep}} = \mathrm{diag}(\boldsymbol{M}_1, \dots, \boldsymbol{M}_m)$, with

$$\boldsymbol{M}_j = \boldsymbol{I} - \boldsymbol{L}_j\boldsymbol{C}, \qquad \boldsymbol{L}_j = \boldsymbol{W}_{jj}\boldsymbol{C}'(\boldsymbol{C}\boldsymbol{W}_{jj}\boldsymbol{C}')^{-1}. \tag{6}$$

Both maps have a regression interpretation. The incoherence of the base forecasts, $\boldsymbol{C}_*\widehat{\boldsymbol{y}} = -\boldsymbol{C}_*\boldsymbol{e}$, is observable, and $\boldsymbol{L}$ is the matrix of population regression coefficients of the errors $\boldsymbol{e}$ on the incoherences $\boldsymbol{C}_*\boldsymbol{e}$, since $\mathrm{Cov}(\boldsymbol{e}, \boldsymbol{C}_*\boldsymbol{e}) = \boldsymbol{W}\boldsymbol{C}_*'$. MinT adds the fitted value of this regression, an estimate of the base forecast error computed from the observed incoherences, to the base forecasts: $\widetilde{\boldsymbol{y}} = \widehat{\boldsymbol{y}} + \boldsymbol{L}\boldsymbol{C}_*\boldsymbol{e}$. Separate reconciliation runs the same regression for each variable, using only its own incoherences $\boldsymbol{C}\boldsymbol{e}_j$. The off-diagonal blocks of $\boldsymbol{L}$ are the coefficients on the other variables' incoherences, and joint reconciliation can only improve on separate reconciliation through them.

3.2. When joint and separate reconciliation coincide

Proposition 1. *Let $\boldsymbol{W}$ be positive definite. The following statements are equivalent.*

(a) $\boldsymbol{M} = \boldsymbol{M}^{\text{sep}}$*: joint and separate reconciliation give the same forecasts for every base forecast.*
(b) $\boldsymbol{M}$ *is block diagonal by variable.*
(c) $\boldsymbol{L}$ *is block diagonal by variable.*
(d) $\boldsymbol{M}_j\boldsymbol{W}_{jk}\boldsymbol{C}' = \boldsymbol{0}$ *for all $j \neq k$; that is,* $\mathrm{Cov}(\boldsymbol{M}_j\boldsymbol{e}_j, \boldsymbol{C}\boldsymbol{e}_k) = \boldsymbol{0}$.
(e) *The column space of $\boldsymbol{W}_{jk}\boldsymbol{C}'$ is contained in that of $\boldsymbol{W}_{jj}\boldsymbol{C}'$ for all $j \neq k$.*

Proof. Let $\boldsymbol{Q} = \boldsymbol{C}_*\boldsymbol{W}\boldsymbol{C}_*'$, with blocks $\boldsymbol{Q}_{jk} = \boldsymbol{C}\boldsymbol{W}_{jk}\boldsymbol{C}'$. Because $\boldsymbol{W}$ is positive definite and $\boldsymbol{C}_*$ has full row rank, $\boldsymbol{Q}$ is invertible, so $\boldsymbol{L}$ is the unique solution of $\boldsymbol{L}\boldsymbol{Q} = \boldsymbol{W}\boldsymbol{C}_*'$. Block (j,k) of the right-hand side is $\boldsymbol{W}_{jk}\boldsymbol{C}'$.

(c)⇒(a). Let $\boldsymbol{L} = \mathrm{diag}(\boldsymbol{H}_1, \dots, \boldsymbol{H}_m)$. Block (j,j) of $\boldsymbol{L}\boldsymbol{Q} = \boldsymbol{W}\boldsymbol{C}_*'$ gives $\boldsymbol{H}_j\boldsymbol{C}\boldsymbol{W}_{jj}\boldsymbol{C}' = \boldsymbol{W}_{jj}\boldsymbol{C}'$, so $\boldsymbol{H}_j = \boldsymbol{L}_j$. Hence $\boldsymbol{M} = \boldsymbol{I} - \boldsymbol{L}\boldsymbol{C}_* = \mathrm{diag}(\boldsymbol{I} - \boldsymbol{L}_j\boldsymbol{C}) = \boldsymbol{M}^{\text{sep}}$.

(a)⇒(b) is immediate.

(b)⇒(c). $\boldsymbol{L}\boldsymbol{C}_* = \boldsymbol{I} - \boldsymbol{M}$ is block diagonal. Since $\boldsymbol{C}_*$ is block diagonal with full row rank, it has the block diagonal right inverse $\boldsymbol{C}_*'(\boldsymbol{C}_*\boldsymbol{C}_*')^{-1}$, and $\boldsymbol{L} = (\boldsymbol{L}\boldsymbol{C}_*)\boldsymbol{C}_*'(\boldsymbol{C}_*\boldsymbol{C}_*')^{-1}$ is block diagonal.

(c)⇔(d). By the argument for (c)⇒(a), $\boldsymbol{L}$ is block diagonal if and only if $\boldsymbol{L} = \boldsymbol{L}^{\text{sep}} = \mathrm{diag}(\boldsymbol{L}_1, \dots, \boldsymbol{L}_m)$. By uniqueness, this holds if and only if $\boldsymbol{L}^{\text{sep}}\boldsymbol{Q} = \boldsymbol{W}\boldsymbol{C}_*'$. Blocks (j,j) of this equation hold by the definition of $\boldsymbol{L}_j$. Blocks (j,k) with $j \neq k$ read $\boldsymbol{L}_j\boldsymbol{C}\boldsymbol{W}_{jk}\boldsymbol{C}' = \boldsymbol{W}_{jk}\boldsymbol{C}'$, that is, $(\boldsymbol{I} - \boldsymbol{L}_j\boldsymbol{C})\boldsymbol{W}_{jk}\boldsymbol{C}' = \boldsymbol{M}_j\boldsymbol{W}_{jk}\boldsymbol{C}' = \boldsymbol{0}$. Finally, $\mathrm{Cov}(\boldsymbol{M}_j\boldsymbol{e}_j, \boldsymbol{C}\boldsymbol{e}_k) = \boldsymbol{M}_j\boldsymbol{W}_{jk}\boldsymbol{C}'$.

(d)⇔(e). Condition (d) states $\boldsymbol{W}_{jk}\boldsymbol{C}' = \boldsymbol{W}_{jj}\boldsymbol{C}'\boldsymbol{D}_{jk}$ with $\boldsymbol{D}_{jk} = (\boldsymbol{C}\boldsymbol{W}_{jj}\boldsymbol{C}')^{-1}\boldsymbol{C}\boldsymbol{W}_{jk}\boldsymbol{C}'$, so (d) implies (e). Conversely, if $\boldsymbol{W}_{jk}\boldsymbol{C}' = \boldsymbol{W}_{jj}\boldsymbol{C}'\boldsymbol{D}$ for some $\boldsymbol{D}$, then $\boldsymbol{M}_j\boldsymbol{W}_{jk}\boldsymbol{C}' = \boldsymbol{M}_j\boldsymbol{W}_{jj}\boldsymbol{C}'\boldsymbol{D} = \boldsymbol{0}$, because $\boldsymbol{M}_j\boldsymbol{W}_{jj}\boldsymbol{C}' = \boldsymbol{W}_{jj}\boldsymbol{C}' - \boldsymbol{L}_j\boldsymbol{C}\boldsymbol{W}_{jj}\boldsymbol{C}' = \boldsymbol{0}$. □

Condition (d) has a direct reading. Joint reconciliation changes nothing if and only if, after each variable has been reconciled on its own, its remaining errors are uncorrelated with the incoherences of every other variable. Condition (b) shows that there is no intermediate case. If the reconciled forecasts

of one variable depend on the base forecasts of another at all, they differ from the separately reconciled forecasts.

Two known results are special cases of Proposition 1.

Corollary 1 (Invariance)**.** *For any $\boldsymbol{K}$ such that $\boldsymbol{W} + \boldsymbol{S}_*\boldsymbol{K}\boldsymbol{S}_*'$ is positive definite, replacing $\boldsymbol{W}$ by $\boldsymbol{W} + \boldsymbol{S}_*\boldsymbol{K}\boldsymbol{S}_*'$ leaves $\boldsymbol{M}$, $\boldsymbol{M}^{\text{sep}}$ and $\boldsymbol{L}$ unchanged.*

Proof. Since $\boldsymbol{C}\boldsymbol{S} = \boldsymbol{0}$, we have $\boldsymbol{C}_*\boldsymbol{S}_* = \boldsymbol{0}$, so $(\boldsymbol{W} + \boldsymbol{S}_*\boldsymbol{K}\boldsymbol{S}_*')\boldsymbol{C}_*' = \boldsymbol{W}\boldsymbol{C}_*'$ and $\boldsymbol{C}_*(\boldsymbol{W} + \boldsymbol{S}_*\boldsymbol{K}\boldsymbol{S}_*')\boldsymbol{C}_*' = \boldsymbol{C}_*\boldsymbol{W}\boldsymbol{C}_*'$. Hence $\boldsymbol{L}$, and so $\boldsymbol{M}$, is unchanged. The diagonal blocks change by $\boldsymbol{S}\boldsymbol{K}_{jj}\boldsymbol{S}'$, which leaves each $\boldsymbol{L}_j$, and so $\boldsymbol{M}^{\text{sep}}$, unchanged by the same argument. □

This is the argument used by Wickramasuriya (2021, Proposition 1) to show that the GLS reconciliation of Hyndman et al. (2011) and MinT coincide. It is an instance of the classical result that generalised least squares is unaffected by adding $\boldsymbol{X}\boldsymbol{K}\boldsymbol{X}'$ to the error covariance, where $\boldsymbol{X}$ is the regressor matrix (Rao, 1967; Zyskind, 1967). Only the part of the error covariance that is visible through the incoherences matters. Any cross-variable dependence of the form $\boldsymbol{S}\boldsymbol{K}_{jk}\boldsymbol{S}'$ is invisible to both joint and separate reconciliation. The same invariance gives the known condition $\boldsymbol{W} = \boldsymbol{S}_*\boldsymbol{K}\boldsymbol{S}_*' + \tau\boldsymbol{I}$ under which MinT reduces to OLS reconciliation (Hyndman et al., 2011; Wickramasuriya, 2021).

Corollary 2 (Separability)**.** *If $\boldsymbol{W} = \boldsymbol{V} \otimes \boldsymbol{\Sigma}_0$ for positive definite matrices $\boldsymbol{V}$ $(m \times m)$ and $\boldsymbol{\Sigma}_0$ $(n \times n)$, then $\boldsymbol{M} = \boldsymbol{M}^{\text{sep}} = \boldsymbol{I}_m \otimes \boldsymbol{M}_0$, where $\boldsymbol{M}_0 = \boldsymbol{I} - \boldsymbol{\Sigma}_0\boldsymbol{C}'(\boldsymbol{C}\boldsymbol{\Sigma}_0\boldsymbol{C}')^{-1}\boldsymbol{C}$, whatever the sign or strength of the cross-variable dependence in $\boldsymbol{V}$.*

Proof. Here $\boldsymbol{W}_{jk} = v_{jk}\boldsymbol{\Sigma}_0$, so $\boldsymbol{L}_j = \boldsymbol{\Sigma}_0\boldsymbol{C}'(\boldsymbol{C}\boldsymbol{\Sigma}_0\boldsymbol{C}')^{-1}$ and $\boldsymbol{M}_j = \boldsymbol{M}_0$ for every j. Then $\boldsymbol{W}_{jk}\boldsymbol{C}' = (v_{jk}/v_{jj})\boldsymbol{W}_{jj}\boldsymbol{C}'$, and condition (e) holds. □

Corollary 2 is a special case of a result of Girolimetto and Di Fonzo (2025) for cross-temporal reconciliation in which the second dimension carries no aggregation constraints. It is the reconciliation analogue of the result that seemingly unrelated regressions with identical regressors reduce to equation-by-equation least squares (Zellner, 1962): every variable shares the same summing matrix, so cross-variable correlation alone cannot be exploited. Corollaries 1 and 2 combine: every $\boldsymbol{W} = \boldsymbol{V} \otimes \boldsymbol{\Sigma}_0 + \boldsymbol{S}_*\boldsymbol{K}\boldsymbol{S}_*'$ satisfies the conditions of Proposition 1, however far it is from a Kronecker product. So separability is sufficient but far from necessary.

This limits the value of measuring separability directly. If $\boldsymbol{W}$ is rearranged so that each block $\boldsymbol{W}_{jk}$ becomes a row, it is a Kronecker product if and only if the rearranged matrix has rank one, and the relative error of the nearest Kronecker approximation (Van Loan and Pitsianis, 1993) is $\{1 - \sigma_1^2/\sum_i \sigma_i^2\}^{1/2}$, where σ_i are the singular values of the rearranged matrix. Likelihood ratio tests of separability are also available (Lu and Zimmerman, 2005; Mitchell et al., 2006). Both measures of separability respond to components $\boldsymbol{S}_*\boldsymbol{K}\boldsymbol{S}_*'$ that do not change the reconciled forecasts, and a departure that does change them may leave the Kronecker error small.

3.3. Remarks

Horizon. Under the usual simplifying assumption $\boldsymbol{W}_h = k_h\boldsymbol{W}_1$, the maps $\boldsymbol{M}$ and $\boldsymbol{M}^{\text{sep}}$ do not depend on h, and neither do the conditions of Proposition 1.

Shrinkage. The diagonal target of the shrinkage estimator has zero cross-variable blocks and so satisfies condition (d). Full shrinkage therefore makes joint reconciliation identical to reconciling each variable separately with its own diagonal target, and partial shrinkage scales the estimated cross-variable blocks of $\boldsymbol{W}$ towards zero.

Linear combinations. MinT minimises not only the trace of the reconciled error covariance but the whole matrix in the positive semidefinite order, since it is the generalised least squares projection (Wickramasuriya et al., 2019). Separate reconciliation is another linear reconciliation of this form (its $\boldsymbol{G}$ satisfies $\boldsymbol{G}\boldsymbol{S}_* = \boldsymbol{I}$), so for every linear combination $\boldsymbol{\ell}'\boldsymbol{y}$ of the stacked series, such as the difference between two variables at one node, joint reconciliation with the true $\boldsymbol{W}$ has no larger error variance.

When the conditions of Proposition 1 fail, the gain need not be spread evenly: it can be negligible for each variable's own series yet substantial for combinations that load on several variables.

More than two variables. Proposition 1 holds for any m, and condition (d) can be checked pair by pair. Joint reconciliation of m variables can be split into separate reconciliations of groups of variables whenever the off-diagonal blocks of $\boldsymbol{L}$ between groups vanish.

4. Measuring whether joint reconciliation can help

Proposition 1 gives an exact condition, but in practice $\boldsymbol{W}$ is estimated and the condition never holds exactly in the estimate. We need a measure of how far an estimated covariance is from satisfying it, and a way to judge whether the observed departure is larger than estimation noise alone would produce.

4.1. Two measures

We measure departures from the conditions of Proposition 1 directly, rather than the distance from a Kronecker product (Section 3). The first measure describes the structure of the joint map. Since $\boldsymbol{L}$ contains the regression coefficients of the base forecast errors on the incoherences, and its off-diagonal blocks are the coefficients on other variables' incoherences, we define

$$\kappa(\boldsymbol{W}) = \left(\frac{\sum_{j \neq k} \|\boldsymbol{L}_{jk}\|_F^2}{\|\boldsymbol{L}\|_F^2} \right)^{1/2}, \tag{7}$$

where $\boldsymbol{L}_{jk}$ is the $(n \times n_a)$ block of $\boldsymbol{L}$ relating the errors of variable j to the incoherences of variable k. Rescaling variable k by c_k multiplies $\boldsymbol{L}_{jk}$ by c_j / c_k, so before computing κ we standardise each variable by the square root of the mean of the diagonal of $\boldsymbol{C}\boldsymbol{W}_{jj}\boldsymbol{C}'$ (the variances of its incoherences). This makes κ free of units. By Proposition 1, $\kappa = 0$ if and only if joint and separate reconciliation coincide. Adding a component $\boldsymbol{S}_*\boldsymbol{K}\boldsymbol{S}_*'$ to $\boldsymbol{W}$ leaves $\boldsymbol{L}$ unchanged (Corollary 1), and since $\boldsymbol{C}\boldsymbol{S} = \boldsymbol{0}$ it leaves the standardisation unchanged too, so the value of κ does not depend on such components.

The second measure describes the value of the joint map. If $\boldsymbol{W}$ were the true error covariance, the proportional reduction in mean squared error (MSE) from reconciling jointly rather than separately would be

$$\gamma(\boldsymbol{W}) = 1 - \frac{\operatorname{tr}(\boldsymbol{M}\boldsymbol{W}\boldsymbol{M}')}{\operatorname{tr}(\boldsymbol{M}^{\text{sep}}\boldsymbol{W}\boldsymbol{M}^{\text{sep}\prime})}. \tag{8}$$

MinT minimises the trace over all maps of this form, so $\gamma \geq 0$, again with equality if and only if the conditions of Proposition 1 hold. Positive semidefinite components $\boldsymbol{S}_*\boldsymbol{K}\boldsymbol{S}_*'$ pass through both maps unchanged and add the same amount to both traces, so they shrink γ towards zero without changing whether it is zero: γ is the gain relative to the whole error, including the part that no reconciliation can remove. Unlike κ, γ depends on the size of the incoherent component of the errors, because the gain from any reconciliation comes only from that component. So γ is small whenever that component is small, however strongly the joint map leans on other variables' incoherences. A large κ with a small γ says that joint reconciliation would change the forecasts but not improve them by much on average. Because γ averages over all series, it can hide a larger gain for a particular combination of variables. For a combination $\boldsymbol{\ell}'\boldsymbol{y}$ of interest we therefore also compute

$$\gamma_{\boldsymbol{\ell}}(\boldsymbol{W}) = 1 - \frac{\boldsymbol{\ell}'\boldsymbol{M}\boldsymbol{W}\boldsymbol{M}'\boldsymbol{\ell}}{\boldsymbol{\ell}'\boldsymbol{M}^{\text{sep}}\boldsymbol{W}\boldsymbol{M}^{\text{sep}\prime}\boldsymbol{\ell}'}, \tag{9}$$

which is also non-negative for every $\boldsymbol{\ell}$ (Section 3).

4.2. Estimation and testing

Both measures are computed from an estimate $\widehat{\boldsymbol{W}}$, and both are biased in finite samples. Sampling noise pushes them above zero even when the population covariance satisfies Proposition 1. The shrinkage estimator pulls the other way, because its diagonal target has no cross-variable blocks

(Section 3), so it shrinks genuine departures towards zero. So a non-zero $\widehat{\kappa}$ may be pure noise, and a genuine departure may be understated. The estimates $\widehat{\kappa}$ and $\widehat{\gamma}$ are therefore useful as descriptions of what joint reconciliation would do with the estimated covariance, but their raw values cannot be read as evidence that the population conditions fail. For that we need a test.

A direct test of the conditions follows from the regression reading of Proposition 1. Given its own incoherences $\boldsymbol{z}_j = \boldsymbol{C}\boldsymbol{e}_j$, the errors of variable j are determined by their bottom-level part $\boldsymbol{e}_j^b$, since the aggregate errors equal $\boldsymbol{z}_j + \boldsymbol{A}\boldsymbol{e}_j^b$. Condition (d) says that, in the multivariate regression

$$\boldsymbol{e}_{j,t}^b = \boldsymbol{\alpha}_j + \boldsymbol{\Theta}_{jj}'\boldsymbol{z}_{j,t} + \sum_{k \neq j} \boldsymbol{\Theta}_{jk}'\boldsymbol{z}_{k,t} + \boldsymbol{\xi}_{j,t}, \tag{10}$$

the other variables' incoherences add nothing: $\boldsymbol{\Theta}_{jk} = \boldsymbol{0}$ for all $k \neq j$. This is a standard linear hypothesis in a multivariate regression with n_b responses and $(m-1)n_a$ restrictions per response. We test it with the likelihood ratio statistic, Wilks' Λ, using Rao's F approximation to its null distribution (Rao, 1951), applied to the one-step residuals of the base models. With m variables there are m such tests, which we combine with a Bonferroni correction. The test uses the sample covariance, so it is unaffected by the shrinkage that biases $\widehat{\kappa}$ and $\widehat{\gamma}$. It needs $T \geq n_b + mn_a + 1$, and its null distribution assumes independent residual vectors, so it should be applied to the residuals of base models that leave no serial correlation.

We also tried calibrating $\widehat{\kappa}$ and $\widehat{\gamma}$ by bootstrap, generating null data from a covariance that satisfies Proposition 1 and is fitted to the data. These procedures are badly mis-sized when the number of series is large relative to T, because any null covariance built from $\widehat{\boldsymbol{W}}$ inherits its estimation error. The regression test avoids simulation altogether.

5. Simulations

We use two experiments. The first simulates base forecast errors directly, so that the population error covariance is known and the gain from joint reconciliation can be computed exactly. It also shows how the measures and the test of Section 4 behave when the covariance is estimated. The second follows the usual design of simulation studies of reconciliation: it simulates correlated bottom-level series, aggregates them, fits base forecasting models to every series, and reconciles their forecasts.

5.1. Experiment 1: controlled error covariances

For each of two variables, the base forecast errors are $\boldsymbol{e}_j = \boldsymbol{S}\boldsymbol{u}_j + \boldsymbol{v}_j$. The coherent component $\boldsymbol{S}\boldsymbol{u}_j$ is invisible to reconciliation (Corollary 1). The incoherent component $\boldsymbol{v}_j$ has independent entries across series, with variance ψ_{ij} for series i and correlation ρ_i between the two variables at series i. With this structure, and because the hierarchy has a single total, condition (e) of Proposition 1 holds for both pairs of variables if and only if $\rho_i(\psi_{i2}/\psi_{i1})^{1/2}$ and $\rho_i(\psi_{i1}/\psi_{i2})^{1/2}$ are each the same for every series; that is, when ρ_i and ψ_{i2}/ψ_{i1} are both constant across series, or $\rho_i = 0$ for all i. We consider five families of covariance matrices, each indexed by a parameter: ρ for F0, the size of $\boldsymbol{K}$ for F1 and F4, and δ for F2 and F3. Let s_i be the number of bottom-level series summed by series i (the ith row sum of $\boldsymbol{S}$).

F0, separable. $\psi_{i1} = \psi_{i2} = s_i$ and $\rho_i = \rho$, with ρ from -0.9 to 0.9.

F1, separable plus invisible. F0 with $\rho = 0.7$, plus a non-separable coherent component $\boldsymbol{S}_*\boldsymbol{K}\boldsymbol{S}_*'$ of increasing size.

F2, variable-specific profiles. $\psi_{i1} = s_i$, $\psi_{i2} = s_i^{1+\delta}$ and $\rho_i = 0.7$. As δ grows, the second variable's incoherent errors become increasingly concentrated at the aggregate levels.

F3, node-varying correlation. $\psi_{i1} = \psi_{i2} = s_i$ and $\rho_i = 0.3 + \delta\omega_i$, where the ω_i are fixed values in $[-1, 1]$, spread evenly and assigned to the series in a scrambled order.

F4, OLS case. $\boldsymbol{W} = \boldsymbol{V} \otimes \boldsymbol{I}_n + \boldsymbol{S}_*\boldsymbol{K}\boldsymbol{S}_*'$, with unit variances and cross-variable correlation 0.7 in $\boldsymbol{V}$, so that separate reconciliation of each variable is OLS reconciliation.

F0, F1 and F4 satisfy Proposition 1; F2 and F3 do not when $\delta > 0$. We use two hierarchies: the eight-series hierarchy of Figure 1, and a 33-series hierarchy in which 27 bottom-level series aggregate

to five intermediate series and then to a single total. The second has the structure of the Brazilian employment data analysed in Section 6, and we refer to it as the Brazilian hierarchy.

Population results.

Figure 2 shows the exact MSE reduction from joint rather than separate reconciliation, $\gamma(\boldsymbol{W})$, against $\kappa(\boldsymbol{W})$. For F0, F1 and F4 both measures are zero to numerical precision, whatever the strength of cross-variable correlation, and for F4 joint MinT coincides with OLS reconciliation. For F2 and F3 the gain rises with κ, but it is moderate: at most 15.7% for variable-specific profiles in the Brazilian hierarchy, where κ reaches 0.54, and at most 5.9% for node-varying correlations in the small hierarchy. Departures from the conditions of Proposition 1 have to be large before joint reconciliation gains more than a few per cent.

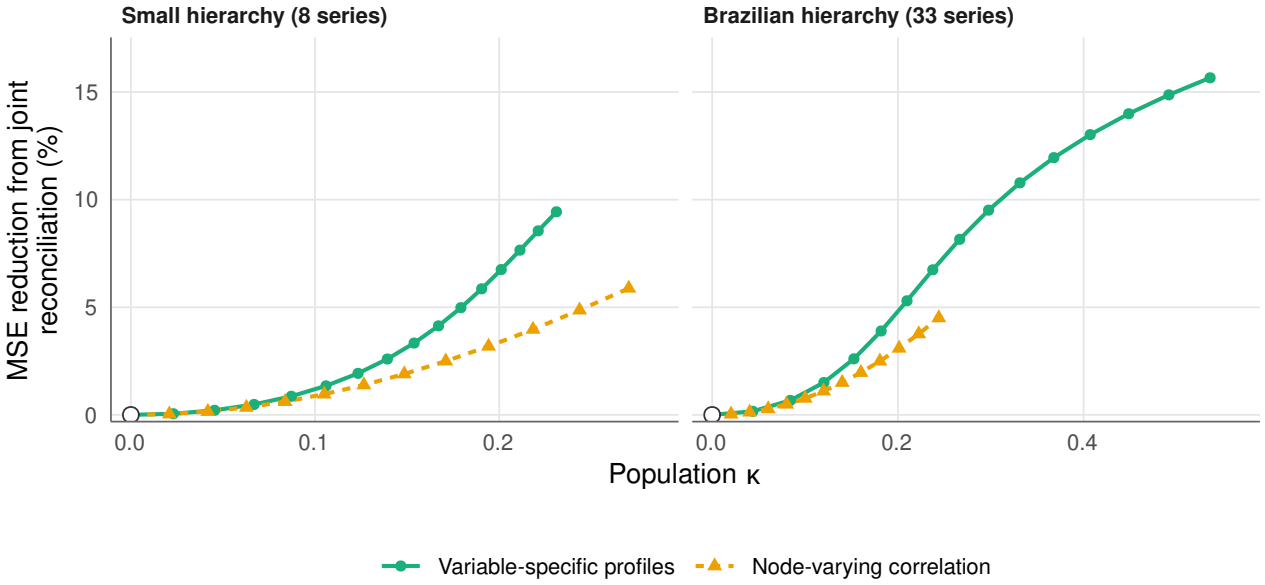


Figure 2: Experiment 1: population MSE reduction from joint rather than separate reconciliation against κ. The separable (F0), separable-plus-invisible (F1) and OLS (F4) families all lie at the origin.

Figure 3 shows why distance from a Kronecker product is the wrong measure (Section 3). It plots the same population gains against the relative error of the nearest Kronecker approximation to $\boldsymbol{W}$. For F1 the Kronecker error rises with no gain at all, because the departure is invisible to reconciliation, and for F2 it can fall while the gain rises.

Estimated covariances.

In practice $\boldsymbol{W}$ is estimated, and the joint estimate has $m^2 = 4$ times as many entries as each separate one. For each family and hierarchy we drew $T \in \{50, 100, 200, 500, 1000, 2000\}$ error vectors from $N(\mathbf{0}, \boldsymbol{W})$, estimated the covariance by shrinkage (4), and computed the exact expected loss $\operatorname{tr}(\widehat{\boldsymbol{M}}\boldsymbol{W}\widehat{\boldsymbol{M}}')$ of each estimated map $\widehat{\boldsymbol{M}}$, averaged over 1000 replications. Figure 4 shows the ratio of expected loss for joint to separate reconciliation. Under separability the joint map is never better, and it is worse at small sample sizes: by 2.9% at $T = 50$ in the small hierarchy and 1.6% in the Brazilian one. That is the price of estimating cross-variable blocks that carry no information. When the population gain is small, the estimated joint map does not recover it until T is in the hundreds or thousands; for the weakest node-varying departure in the Brazilian hierarchy, whose population gain is 0.5%, the estimated joint map gains only 0.1% even at $T = 2000$. Only the largest departures give a clear gain at sample sizes typical of monthly data.

The same replications show the finite-sample behaviour of $\widehat{\kappa}$ (Figure 5). Under a separable covariance the median of $\widehat{\kappa}$ is about 0.12 at sample sizes typical of monthly data and falls only slowly with T. For the strongest node-varying departure (F3) in the Brazilian hierarchy, $\widehat{\kappa}$ is below its population value at

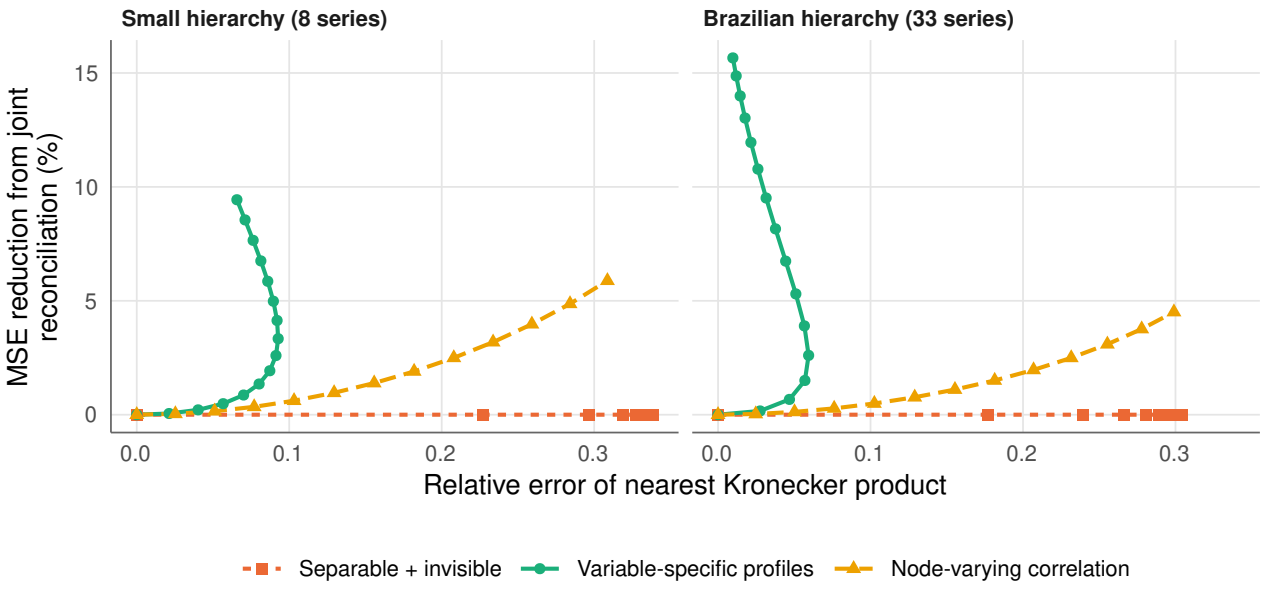


Figure 3: Experiment 1: population MSE reduction from joint rather than separate reconciliation against the relative error of the nearest Kronecker approximation to $\boldsymbol{W}$. Each line joins the points for one family in order of its parameter, starting from the origin, where $\boldsymbol{W}$ is separable. For F1 the Kronecker error grows with the size of the invisible component while the gain stays at zero. For F2 the gain rises throughout, but the Kronecker error rises and then falls as δ increases, so the line doubles back.

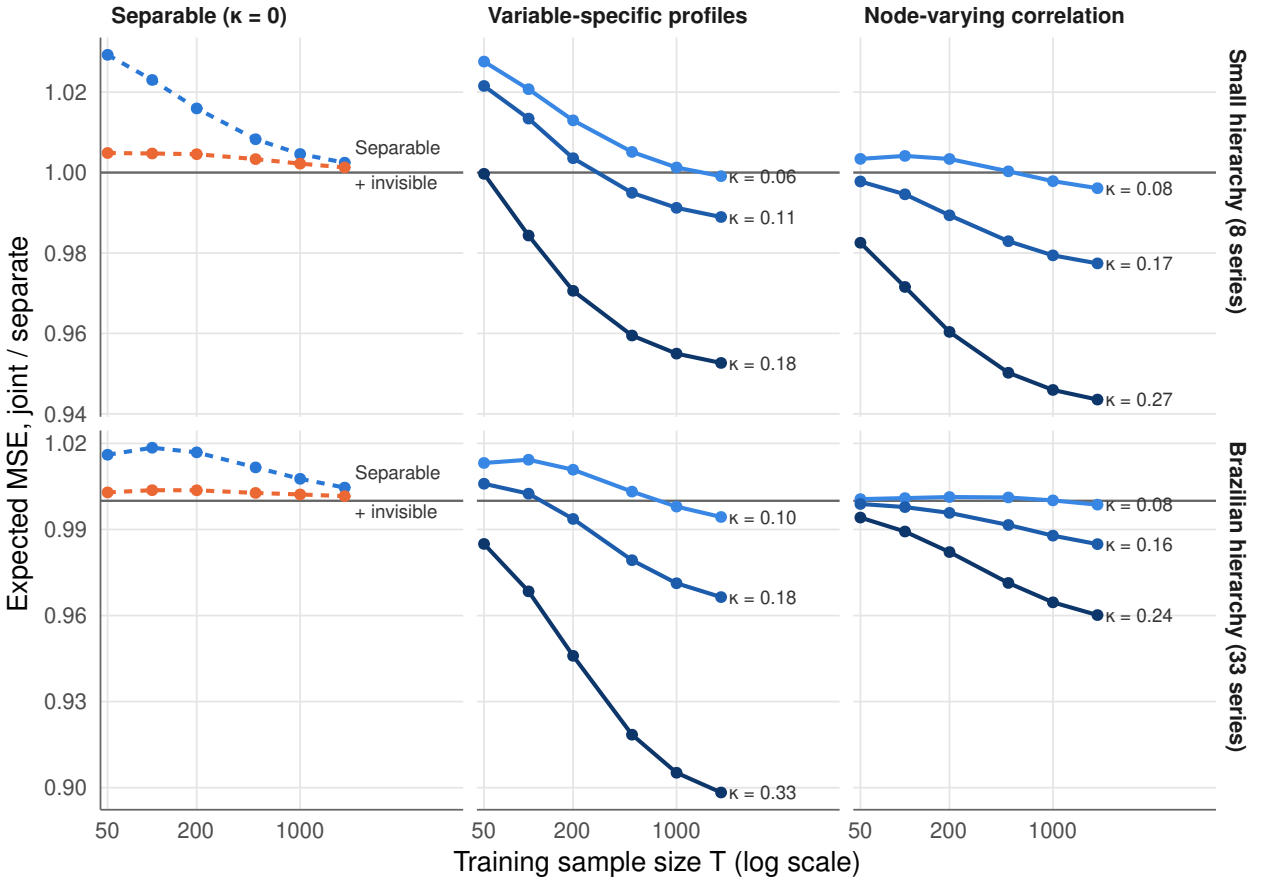


Figure 4: Experiment 1: expected MSE of joint relative to separate reconciliation with shrinkage estimates of $\boldsymbol{W}$, by training sample size. Values below one favour joint reconciliation. Labels give the population κ.

every sample size, because of shrinkage, and at $T \leq 200$ it is about the same as, or smaller than, its value under separability. The raw value of $\widehat{\kappa}$ cannot distinguish the two.

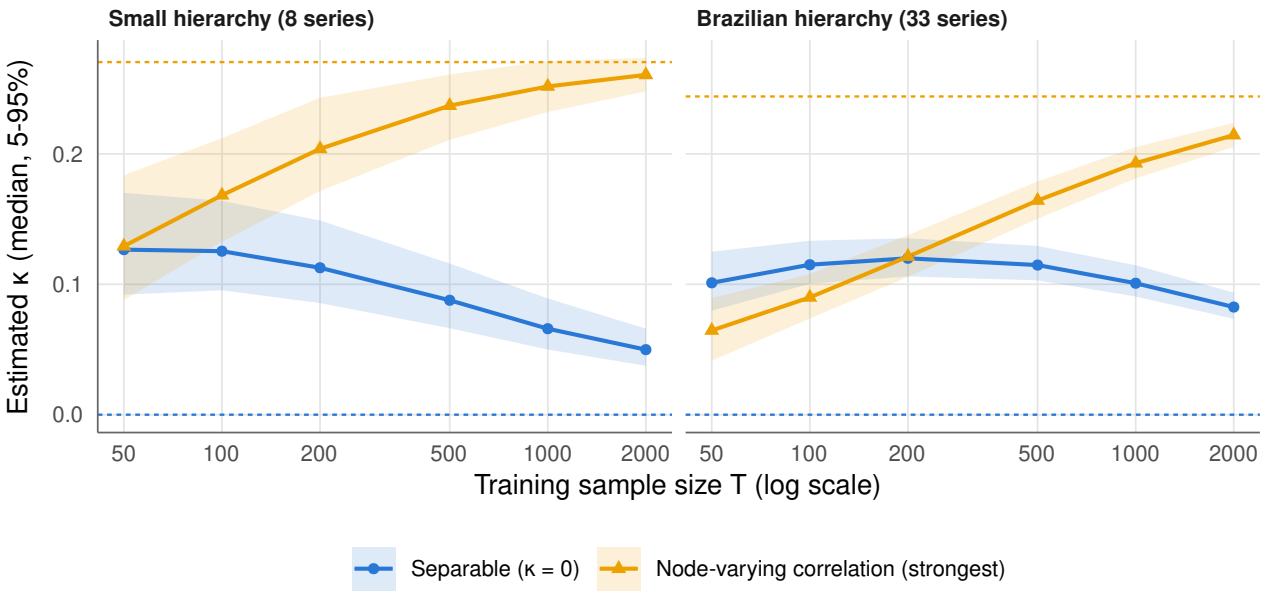


Figure 5: Experiment 1: sampling distribution of $\hat{\kappa}$ (median and 5–95% range) under the separable family F0 and under the node-varying family F3 with the largest departure, with the population values dashed.

Size and power of the test.

To assess the test of Section 4, we drew $T \in \{100, 200, 500, 1000\}$ error vectors from $N(\mathbf{0}, \boldsymbol{W})$ for F0, F1 and three strengths each of F2 and F3, in both hierarchies, and applied the test to each sample (1000 replications). Figure 6 shows the results. Under both separable and separable-plus-invisible covariances, the rejection rate at the 5% level lies between 4.2% and 5.8% across all sample sizes and both hierarchies. Power grows with the size of the departure and with T. Departures with population gains below 1% are detected in at most 30% of samples of 100 or 200 observations, the lengths of typical monthly series, and the smallest of them, a gain of 0.3, 0.3, 0.3, 0.3%, is detected only 50% of the time even with $T = 1000$.

5.2. *Experiment 2: fitted base models*

The second experiment simulates time series and fits base models to them. The two bottom-level series at node i are $\boldsymbol{x}_{i,t} = \boldsymbol{\mu}_{i,t} + \boldsymbol{\eta}_{i,t}$, with a deterministic quarterly component $\boldsymbol{\mu}_{i,t}$ and a bivariate VAR(1) $\boldsymbol{\eta}_{i,t} = \boldsymbol{\Phi}_i \boldsymbol{\eta}_{i,t-1} + \boldsymbol{\varepsilon}_{i,t}$, aggregated over the eight-series hierarchy of Figure 1. We use six scenarios. The three controls have the same dynamics at every node and innovation covariances of the form $\mathrm{Cov}(\boldsymbol{\varepsilon}_{i,t}, \boldsymbol{\varepsilon}_{i',t}) = \sigma_{ii'}\boldsymbol{R}$; they differ in having cross-variable correlation 0.7 in $\boldsymbol{R}$, correlation -0.7, or variable-specific dynamics $\boldsymbol{\Phi}_i = \mathrm{diag}(0.9, 0.2)$. The other three scenarios break this common structure across nodes in a way that differs between the variables: a cross-variable innovation correlation that varies across nodes from -0.6 to 0.9; a different node covariance for each variable; and dynamics that vary across nodes in opposite directions for the two variables.

The controls are a degenerate case. When every node has the same dynamics and the innovation covariances have this form, the autocovariances of every aggregate are proportional to those of the bottom-level series, so the best linear predictor of each series from its own past applies the same filter at every node. A linear filter commutes with aggregation, so the population forecasts are coherent, and any gain from reconciliation, joint or separate, comes from estimation error in the fitted models. When this proportionality fails, as in the other three scenarios and in most simulation designs, a univariate forecast of an aggregate uses only the aggregate's own past, which carries less information than the pasts of its components, so even the population forecasts are incoherent.

For each scenario we computed the one-step error covariance of the population-optimal univariate linear forecasts of all 16 series from AR(20) projections on the exact autocovariances (the results are unchanged to four digits with 40 lags). Table 1 gives κ, γ and the size of the incoherent component,

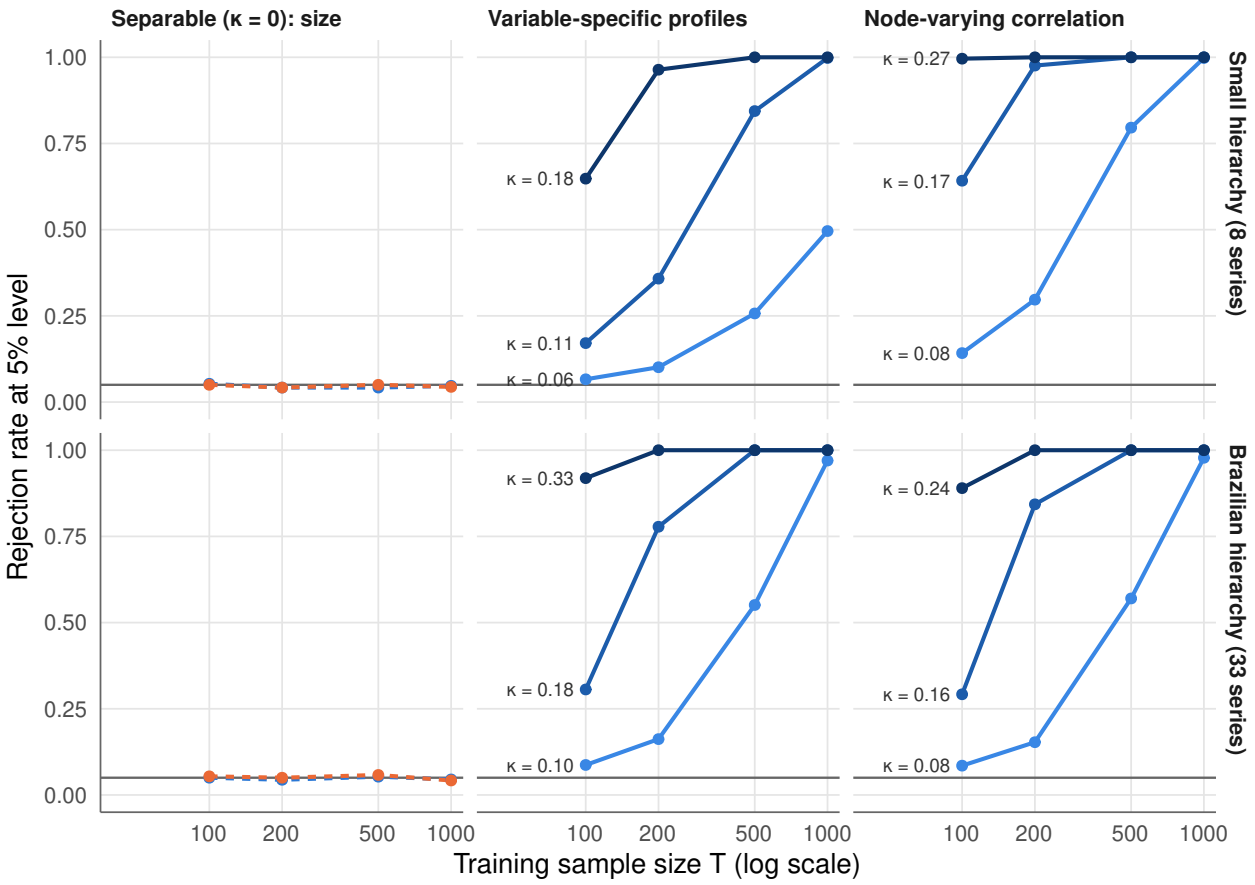


Figure 6: Experiment 1: rejection rate of the test of condition (d) at the 5% level, by sample size (1000 replications). For the separable (F0) and separable-plus-invisible (F1) covariances the rejection rate is the size of the test.

measured by the ratio $\mathrm{tr}(\boldsymbol{C}_*\boldsymbol{W}\boldsymbol{C}_*')\,/\,\mathrm{tr}(\boldsymbol{W})$ of the total variance of the incoherences to that of the errors. The population forecasts in the controls are coherent to machine precision. In the other scenarios this ratio is only 0.4–1.5%. So although κ is large (0.65–0.90), the population gain from joint reconciliation is only 0.5–2.4%.

We then simulated 500 replications of each scenario, with series of length $T + 12$ for $T = 108$ and $T = 400$, fitted automatically selected ARIMA models to every series and a bivariate VAR at every node, and reconciled their forecasts for horizons 1–12. Reconciliation improves on the ARIMA base forecasts (by 6–11% in MSE), but joint and separate reconciliation are nearly indistinguishable. With $T = 108$ the ratio of joint to separate MSE lies in the range 0.996–1.006 for ARIMA base forecasts, and with VAR base forecasts the two never differ by more than 0.5%.

With $T = 400$ and ARIMA base forecasts, joint reconciliation is no worse in any scenario and is better by up to 2.1% in the controls and up to 1.5% in the other three scenarios, so the gain is no smaller in the controls, where the population forecasts are coherent. The fitted models differ from node to node and are estimated from correlated data, so their forecasts are incoherent even in the controls, and the cross-variable structure of this estimation error need not satisfy Proposition 1. Using the diagonal blocks of the joint covariance estimate for separate reconciliation changes the MSE by at most 0.2%, so the gain comes from the cross-variable blocks rather than from differences in shrinkage. In these designs, joint reconciliation exploits the structure of the base models' estimation error rather than that of the data-generating process, and the resulting gains are small.

Table 1: Experiment 2. Population quantities for the optimal AR forecasts: incoherence variance ratio $\mathrm{tr}(\boldsymbol{C}_*\boldsymbol{W}\boldsymbol{C}_*')\,/\,\mathrm{tr}(\boldsymbol{W})$ (%), κ, and MSE reduction from joint reconciliation (%). Fitted base models: MSE of joint relative to separate reconciliation, averaged over 500 replications, 16 series and horizons 1–12, with the standard error of the log ratio in parentheses; mean plug-in gain $\widehat{\gamma}$ (%, ARIMA).

		Population			Joint / separate		
Scenario	T	Incoh.	κ	Gain	ARIMA	VAR	$\widehat{\gamma}$
Separable, $\rho = 0.7$	108	0.00	0.00	0.00	0.997 (0.003)	1.002 (0.001)	1.7
Separable, $\rho = 0.7$	400	0.00	0.00	0.00	0.979 (0.004)	0.999 (0.001)	1.8
Separable, $\rho = -0.7$	108	0.00	0.00	0.00	0.996 (0.003)	1.004 (0.001)	2.1
Separable, $\rho = -0.7$	400	0.00	0.00	0.00	0.987 (0.004)	0.999 (0.001)	2.3
Variable-specific dynamics	108	0.00	0.00	0.00	1.006 (0.002)	1.000 (0.001)	1.8
Variable-specific dynamics	400	0.00	0.00	0.00	0.996 (0.003)	0.997 (0.001)	2.0
Node-varying correlation	108	0.42	0.80	2.14	0.998 (0.003)	1.005 (0.001)	1.5
Node-varying correlation	400	0.42	0.80	2.14	0.991 (0.003)	0.998 (0.001)	1.7
Variable-specific node covariance	108	1.28	0.90	2.37	0.998 (0.003)	1.003 (0.002)	1.6
Variable-specific node covariance	400	1.28	0.90	2.37	0.990 (0.003)	1.001 (0.001)	1.6
Node-varying dynamics	108	1.52	0.65	0.46	0.999 (0.003)	1.000 (0.001)	1.9
Node-varying dynamics	400	1.52	0.65	0.46	0.985 (0.003)	0.995 (0.001)	1.9

6. Application: admissions and dismissals in Brazil

6.1. Data

Brazil's General Register of Employed and Unemployed (CAGED) records every admission and dismissal of formally employed workers. The two variables share a geographical hierarchy: 26 states and the Federal District (27 federative units) aggregate to five regions, which aggregate to the country (Figure 7). So $m = 2$, $n = 33$ and $n_b = 27$, giving 66 series in all.

The Labour Statistics Dissemination Program (PDET) of the Ministry of Labour and Employment publishes monthly counts of admissions and dismissals by federative unit, without seasonal adjustment.

In January 2020 CAGED was replaced by Novo CAGED, which draws on different administrative sources, and the COVID-19 pandemic began soon after. Both events changed the joint behaviour of admissions and dismissals, and would contaminate an estimate of their forecast error covariance. We therefore use only January 2004 to December 2019 (192 months).

Figure 8 shows the national series. Admissions and dismissals move together, with strong seasonality: December brings large net job losses every year. Net employment change was positive until 2014, negative through the recession of 2015–2017, and positive again from 2018.

6.2. Forecasting set-up

We forecast all 66 series from rolling origins with an expanding window. The first origin has 108 months of training data (January 2004 to December 2012), and each later origin adds one month. This gives 84 origins, 73 of which have all twelve horizons within the sample. There are two sets of base forecasts. The first comes from an ARIMA model selected automatically for each series (Hyndman and Athanasopoulos, 2021). The second comes from a bivariate VAR for admissions and dismissals at each node, with no seasonal terms and a lag order of up to five selected by the AICc. Results for ETS base forecasts are in the Supplementary Material. We compare the base forecasts with OLS, structurally weighted least squares (WLS) and MinT reconciliation. MinT is applied in three ways: separately to each variable, jointly to both, and separately using the diagonal blocks of the joint covariance estimate. The third isolates the effect of the cross-variable blocks from that of the different shrinkage intensities of the separate and joint estimates.

6.3. Diagnostics

Table 2 reports diagnostics computed from the one-step residuals of models fitted to the whole of 2004–2019. For the ARIMA base forecasts, the estimated error covariance is close to separable: the relative error of the nearest Kronecker product is 0.07. Joint reconciliation would place moderate



Figure 7: Brazilian regions and federative units.

weight on the other variable's incoherences ($\widehat{\kappa} = 0.51$), but the plug-in gain in mean squared error is only 2.6%. The incoherences account for 17% of the total variance of the one-step errors. The likelihood ratio test rejects the conditions of Proposition 1 ($p < 0.001$), so the departure is small but not pure estimation noise. The test assumes serially uncorrelated residuals, however, and a Ljung–Box test with 24 lags rejects at the 5% level for 36% of the ARIMA residual series, so the p-value is only indicative. The VAR residuals give a larger $\widehat{\kappa}$ (0.64) and larger F statistics. But they are autocorrelated for 98% of the series, which invalidates the test: with no seasonal terms and at most five lags, the per-node VARs cannot capture the annual seasonality.

For net employment change the plug-in gain is larger. With ℓ selecting admissions minus dismissals at a node, $\widehat{\gamma}_\ell$ is 6.6% for the national net change and, on average, 5.6% for the regions and 3.1% for the states.

Table 2: Application: diagnostics from the one-step residuals over 2004–2019. Estimated $\widehat{\kappa}$ and plug-in gain $\widehat{\gamma}$ (%); Rao F statistics for the other variable's incoherences in the regression of each variable's errors (162 and 907 degrees of freedom for ARIMA, 162 and 878 for VAR), and the Bonferroni p-value; incoherence variance ratio $\operatorname{tr}(\boldsymbol{C}_* \boldsymbol{W} \boldsymbol{C}_*') / \operatorname{tr}(\boldsymbol{W})$ (%); relative error of the nearest Kronecker product.

				F				
Base model	T	$\widehat{\kappa}$	$\widehat{\gamma}$	Adm.	Dis.	p	Incoh.	Kron.
ARIMA	192	0.510	2.59	1.23	1.60	<0.001	16.8	0.074
VAR	187	0.642	2.83	3.24	2.85	<0.001	11.8	0.086

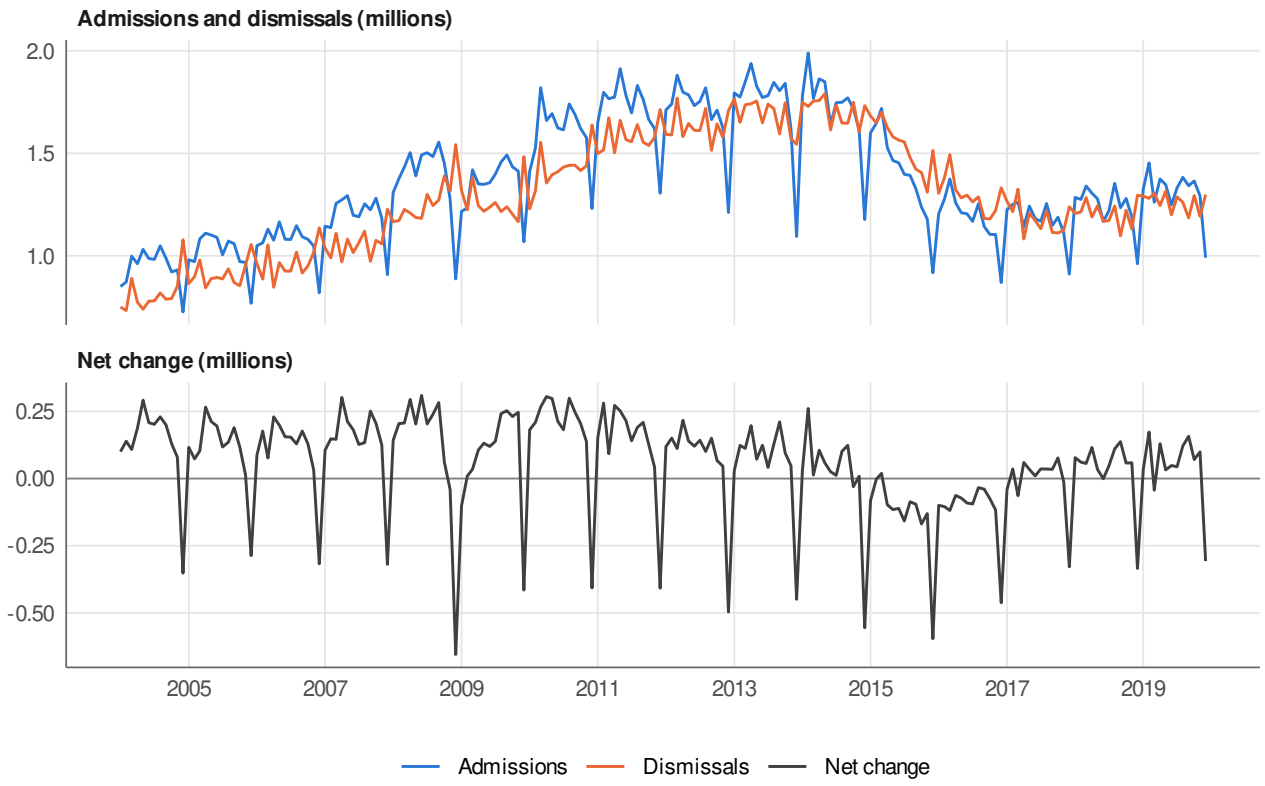


Figure 8: Monthly admissions, dismissals and net employment change in Brazil.

6.4. Forecast accuracy

Table 3 gives the mean squared error of each method relative to the ARIMA base forecasts. For admissions, separate MinT reconciliation improves on the base forecasts at every level. It also beats OLS and structural WLS, which both damage the state-level forecasts.

For dismissals, separate MinT does worse. At the national level its mean squared error is 42% higher than that of the base forecasts, and higher than that of OLS and structural WLS. It is worse than the base forecasts at 67% of the origins, so a few unusual months are not the cause. The reason is that the national ARIMA models forecast dismissals well: summing the state base forecasts gives a mean squared error 30% higher than the national base forecast. MinT shifts weight towards the state-level forecasts, which for dismissals is the wrong direction.

Joint reconciliation recovers part of that loss. For dismissals it beats separate MinT at every level, with ratios of mean squared errors in the range 0.910–0.936, and the gain holds at every horizon (Supplementary Table S3). For admissions it is slightly worse (1.022–1.030). The gain for dismissals is larger than the plug-in gain of 2.6% would suggest.

The results depend on the base model. With VAR base forecasts (Table 4), joint reconciliation is worse than separate reconciliation at every level (ratios 1.010–1.044). The VAR base forecasts are also poorer to begin with: on average over the series, their mean squared error is 35% higher than that of the ARIMA base forecasts. With ETS base forecasts (Supplementary Table S2), joint reconciliation is better for dismissals (0.929–0.960) and neutral for admissions (0.993–1.001). So the benefit for dismissals appears where separate reconciliation is itself poor.

6.5. Net employment change

Net employment change is a linear combination of the two variables, so the argument of Section 3 suggests that joint reconciliation may help it by more than it helps either variable alone. Table 5 shows that it does at the national and regional levels. Compared with separate reconciliation, joint reconciliation reduces the mean squared error of net change by 17% at the national level, 13% for

Table 3: Application: MSE relative to base forecasts (ARIMA), geometric mean over the series at each level, averaged over 84 forecast origins (73 with all twelve horizons) and horizons 1–12.

	Admissions			Dismissals		
Method	Total	Regions	States	Total	Regions	States
Base	1.000	1.000	1.000	1.000	1.000	1.000
OLS	0.989	1.085	1.206	1.033	0.877	1.311
WLS (structural)	0.968	1.025	1.057	1.150	0.864	1.100
MinT, separate	0.952	0.981	0.951	1.420	1.048	1.117
MinT, separate (joint estimate)	0.954	0.987	0.958	1.408	1.040	1.110
MinT, joint	0.981	1.002	0.972	1.302	0.954	1.046

Table 4: Application: MSE relative to base forecasts (VAR), geometric mean over the series at each level, averaged over 84 forecast origins (73 with all twelve horizons) and horizons 1–12.

	Admissions			Dismissals		
Method	Total	Regions	States	Total	Regions	States
Base	1.000	1.000	1.000	1.000	1.000	1.000
OLS	0.994	1.040	1.374	0.991	1.031	1.227
WLS (structural)	0.970	0.976	1.168	0.959	0.967	1.097
MinT, separate	0.888	0.880	0.955	0.907	0.918	1.008
MinT, separate (joint estimate)	0.903	0.894	0.968	0.913	0.919	1.006
MinT, joint	0.926	0.904	0.972	0.947	0.949	1.018

the regions and 6% for the states. It makes the errors in admissions and dismissals more positively correlated, so more of the error cancels in the difference. Pooled over all series and horizons, their correlation is 0.87 after joint reconciliation and 0.83 after separate reconciliation. Almost all of the gain comes from the cross-variable blocks. The different shrinkage intensity of the joint covariance estimate contributes little: separate reconciliation with the joint estimate reduces the mean squared error by at most 1.0%.

For probabilistic forecasts, we generated 1000 sample paths of all 66 series at each origin from the fitted ARIMA models, driven by resampled one-step residuals, and reconciled every path (Panagiotelis et al., 2023). Resampling each variable's innovations independently discards the dependence between admissions and dismissals, while resampling whole residual vectors keeps it. With separate reconciliation, keeping the dependence reduces the continuous ranked probability score (CRPS) of net change by 3% nationally, 2% for the regions and 1% for the states. Reconciling the paths jointly raises the reduction to 12% nationally and 9% for the regions.

Table 5: Application: forecasts of net employment change (admissions minus dismissals), ARIMA base models, geometric mean over the series at each level, 84 origins (73 with all twelve horizons), horizons 1–12. Top: MSE of point forecasts relative to separate MinT. Bottom: CRPS from 1000 sample paths relative to independent innovations with separate reconciliation.

	Total	Regions	States
Point forecasts (MSE)			
MinT, separate	1.000	1.000	1.000
MinT, separate (joint estimate)	0.991	0.990	0.996
MinT, joint	0.832	0.865	0.943
Sample paths (CRPS)			
Independent innovations, separate reconciliation	1.000	1.000	1.000
Joint innovations, separate reconciliation	0.970	0.978	0.987
Joint innovations, joint reconciliation	0.878	0.905	0.956

7. Discussion

When several variables are observed over the same hierarchy and no constraint links them, joint MinT reconciliation can improve on reconciling each variable separately only through the covariance between one variable's reconciled errors and another variable's incoherences. Cross-variable correlation

in the base forecast errors is not enough. Correlation that is common to the whole hierarchy (the separable case) and correlation in the coherent component of the errors are both invisible to reconciliation. What remains is dependence that varies across the hierarchy differently for each variable, and in our simulations and application it is a small fraction of the forecast error variance.

This has three practical consequences.

First, simulation designs need care. The usual design aggregates correlated bottom-level series and forecasts every series on its own. Its population forecasts are incoherent only because a univariate forecast of an aggregate ignores the pasts of its components, and in Experiment 2 these population incoherences are a small fraction of the error variance. When every node shares the same dynamics they vanish altogether, and any difference between joint and separate reconciliation comes from estimation error in the base models. Experiment 2 shows that this estimation error can itself have exploitable cross-variable structure, but that the resulting gains are small and appear only with long series. Varying the cross-variable correlation in such designs, however strongly, says little about the value of joint reconciliation for the data-generating process.

Second, check before reconciling jointly. The measures of Section 4 estimate how much joint reconciliation would change the forecasts (κ) and how much it would improve them (γ), and a likelihood ratio test of condition (d) says whether the departure is distinguishable from estimation noise. When it is not, separate reconciliation is preferable: it does not need the cross-variable blocks of the error covariance, and Experiment 1 shows that estimating those blocks costs accuracy when they carry no information.

Third, think about the quantity being forecast. Many quantities of interest are functions of several variables, such as net employment change. Their forecast distributions depend on the joint distribution of the base forecasts, and in the application, keeping that dependence reduced the CRPS of the national net change by 3%. Because reconciliation is linear, the reconciled joint distribution is the base joint distribution pushed through the reconciliation map. So separate reconciliation of jointly simulated base paths already captures the cross-variable dependence, and it coincides with joint reconciliation whenever the conditions of Proposition 1 hold. When they fail, joint reconciliation can still help such combinations even where it does little for the variables themselves, as it did for net employment change in Brazil. The plug-in gain $\widehat{\gamma}_\ell$ for the combination of interest is a better guide than the average gain $\widehat{\gamma}$.

Joint reconciliation is needed when a constraint does link the variables. If net employment change were forecast as a series in its own right, the constraint that it equals admissions minus dismissals would couple the two variables, and no variable-by-variable reconciliation could satisfy it. Such signed linear constraints fit the general framework of Girolimetto and Di Fonzo (2024). With them $\boldsymbol{C}_*$ is no longer block diagonal, separate reconciliation is not an option, and the question addressed here does not arise.

Several extensions are natural. The shrinkage estimator treats cross-variable covariances like any other off-diagonal entry; an estimator that shrinks towards the set of covariance matrices satisfying Proposition 1 would interpolate between separate and joint reconciliation, with the data choosing how far to move. The measures and the test could also be applied pairwise to decide which groups of variables to reconcile together when there are many. Finally, the same questions arise for temporal and cross-temporal hierarchies with several variables, to which the conditions of Proposition 1 for joint and separate reconciliation to coincide extend directly.

Reproducibility

The data and code needed to reproduce this paper and its supplementary material are available at https://github.com/robjhyndman/mvhts.

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